Note : All the linux commands were ran on online linux terminal

1. Creating and Renaming Files/Directories

Create a directory named test_dir using mkdir.

```
labex@ubuntu:~$ mkdir test-dir
labex@ubuntu:~$ ls
test-dir
labex@ubuntu:~$
```

Inside test_dir, create an empty file called example.txt.

```
labex@ubuntu:~$ cd test-dir
labex@ubuntu:~/test-dir$ touch example.txt
labex@ubuntu:~/test-dir$ ls
example.txt
```

Rename example.txt to renamed_example.txt using mv

```
labex@ubuntu:~/test-dir$ ls
example.txt
labex@ubuntu:~/test-dir$ mv example.txt renamed_example.txt
labex@ubuntu:~/test-dir$ ls
renamed_example.txt
labex@ubuntu:~/test-dir$
```

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2. Viewing File Contents

Use cat to display the contents of /etc/passwd.

```
labex@ubuntu:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
_apt:x:100:65534::/nonexistent:/usr/sbin/nologin
systemd-network:x:101:102:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:102:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
_chrony:x:103:105:Chrony daemon,,,:/var/lib/chrony:/usr/sbin/nologin
sshd:x:104:65534::/run/sshd:/usr/sbin/nologin
user:x:1000:1000:,,,:/home/user:/bin/bash
labex:x:1001:1001::/home/labex/project:/usr/bin/zsh
```

Display only the first 5 lines of /etc/passwd using head.

```
labex@ubuntu:~/test-dir$ head -n 5 /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
labex@ubuntu:~/test-dir$ tail -n 5 /etc/passwd
```

Display only the last 5 lines of /etc/passwd using tail.

```
labex@ubuntu:~/test-dir$ tail -n 5 /etc/passwd
systemd-resolve:x:102:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
_chrony:x:103:105:Chrony daemon,,,:/var/lib/chrony:/usr/sbin/nologin
sshd:x:104:65534::/run/sshd:/usr/sbin/nologin
user:x:1000:1000:,,,:/home/user:/bin/bash
labex:x:1001:1001::/home/labex/project:/usr/bin/zsh
labex@ubuntu:~/test-dir$
```

Note : All the linux commands were ran on online linux terminal

3.Searching for Patterns
Use grep to find all lines containing the word "root" in /etc/passwd.

```
labex@ubuntu:~/test-dir$ cat /etc/passwd | grep root
root:x:0:0:root:/root:/bin/bash
labex@ubuntu:~/test-dir$
```

4. Zipping and Unzipping
Compress the test_dir directory into a file named test_dir.zip using zip.
Unzip test_dir.zip into a new directory named unzipped_dir.

```
$ zip -r test-dir.zip test-dir
  adding: test-dir/ (stored 0%)
$ ls
test-dir  test-dir.zip
$ unzip test-dir.zip -d unzipped_dir
Archive:  test-dir.zip
$    creating: unzipped_dir/test-dir/
$ ls
test-dir  test-dir.zip  unzipped_dir
$
```

5. Downloading Files
Use wget to download a file from a URL (e.g., https://example.com/sample.txt).

wget https://github.com/akku8/GenAI-Honors-Capstone/blob/main/README.md

(Not able to execute this as online linux terminals have less permissions)

Note : All the linux commands were ran on online linux terminal

6. Changing Permissions
Create a file named secure.txt and change its permissions to read-only for everyone using chmod.

```
$ touch secure.txt

$ chmod 444 secure.txt

$ ls -l secure.txt
-r--r--r-- 1 codeuser10006 codeuser10006 0 Mar 18 14:11 secure.txt
$
```

7. Working with Environment Variables
Use export to set a new environment variable called MY_VAR with the value "Hello, Linux!”.

```
$ export MY_VAR="Hello, Linux!"

$ echo $MY_VAR
Hello, Linux!
$
```

Submission Guidelines -: Attach Screenshots or command along with explanation and submit in doc(google doc or microsoft doc) format also attach github repo link

Note : All the linux commands were ran on online linux terminal